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EGYPT GRAND CHALLENGES

- **Recycle garbage and waste for economic and environmental purposes.**
- **Improve the scientific and technological environment for all.**
- **Reduce and adapt to the effect of climatic change.**
- **Increase the industrial and agricultural base of Egypt.**

GRAND CHALLENGES

Recycle garbage and waste for environmental purposes.

solid waste (SW) management is becoming a problem for major cities worldwide. This is especially serious in developing countries as a result of the rapid increase in SW generation caused by rapid population growth, urbanization, rapid industrialization, and economic development. Municipal Solid Waste (MSW) more commonly known as trash or garbage consists of everyday items we use and then throw away, such as product packaging, grass clippings, furniture, clothing, bottles, food scraps, newspapers, appliances, paint, and batteries. This comes from homes, schools, hospitals, and businesses.

In Egypt Municipal solid waste has become a big current problem, the amount of municipal solid waste generated daily is 55 tons, according to estimates by the year 2011, with an estimated amount of waste generated annually of about 20 million tons, and this amount is expected to increase by 2016. Environmental damage not only results from the amount of waste generated but also from the way the wastes are disposed.

Fig 1

Uncollected wastes in the streets cause bad smells, drain blockage, increase the number of scavengers, general public nuisance, and become a good breeding site for insects. Municipal solid waste management (MSWM) involves activities associated with six basic principles of waste generation, storage, collection, transfer and transport, processing, and disposal. However, in most countries, the MSWM system comprises only four activities, i.e., waste generation, collection, transportation, and disposal.

In Egypt, we have a big problem with how we handle our trash. We don't know enough about it, we don't manage it well, and we lack the money, technology, and resources to handle this problem. This leads to problems like throwing trash in the open, not collecting it properly, and burning it without control. To make sure people

are safe and healthy, we need to set up better systems for dealing with our trash. These systems should be good for the environment and make financial sense.

One important part of this is recycling, which means taking things like paper, plastic, and metal from our trash and using them again. Recycling helps save resources, reduces pollution, and can even create jobs. So, in simple terms and

On a large scale, we should recycle our garbage to help the environment and our economy. recycling saves huge amounts of energy that we need for producing new materials. The most important thing here is that we need to promote the culture of recycling; starting from schools, universities, and workplaces to establish effective participation and in turn, have a healthy environment. From my point of view, there are various advantages of recycling such as saving energy, oil resources and reducing pollution.

Improve the scientific and technological environment for all.

The collective panorama of scientific and technological breakthroughs, discoveries, approaches, tools, and infrastructure is referred to as the scientific and technological environment. It includes a range of components, such as:

Scientific Research: This involves the pursuit of knowledge through systematic investigation, experimentation, and analysis in fields such as biology, chemistry, physics, astronomy, etc.
Technological Development: This includes the creation and improvement of tools, devices, processes, and systems aimed at solving problems, improving efficiency, and enhancing human capabilities.
Innovation: The environment fosters innovation, encouraging the transformation of ideas into practical applications or products that benefit society.
Collaboration and Networking: Scientists, engineers, researchers, and technologists often collaborate across disciplines and organizations, sharing knowledge and resources to advance their respective fields.
Regulatory Frameworks: There are often regulations and standards governing scientific research and technological development to ensure safety, ethics, and adherence to legal requirements.
Infrastructure: This encompasses the physical and virtual infrastructure necessary for scientific and technological endeavors, including laboratories, research facilities, computing resources, and communication networks.
Education and Training: A skilled workforce is essential for progress in science and technology, so the environment includes educational institutions, training programs, and initiatives aimed at developing expertise in these fields.
Funding and Investment: Research and development activities require financial resources, so the availability of funding from governments, private organizations, and investors is a crucial aspect of the environment.
Globalization and International Collaboration: Scientific and technological activities often transcend national boundaries, with researchers and organizations collaborating on a global scale.
Ethical and Societal Implications: The environment considers the ethical, social,

and cultural impacts of scientific and technological advancements, promoting responsible innovation and considering the broader implications for society. Scientific information and new developments influence human lives making every human endeavor systematic.

These developments in science and technology lead to educating literate individuals in terms of science and technology, or individuals who can think creatively and deliver their ideas freely. More specifically, individuals in today's world should be educated in a way that they can differentiate causes and effects, analyze ideas, and solve problems efficiently. To educate such individuals, society needs to have a sound grasp of the notions of technology and literacy along with the interrelationship between these notions. Thus, developments in science and technology influence our lives in a gradually increasing way, which leads society to equip future generations with working knowledge of science and technology issues.

Climate change

Climate change is a pressing issue of the 21st century, resulting from human activities like fossil fuel burning and deforestation, leading to global warming. Africa, including Egypt, faces significant threats such as soil degradation, increased pests and diseases, and shifts in agriculture suitability. Egypt's vulnerability is accentuated by its reliance on the Nile River and its coastal exposure to rising sea levels. This study examines Egypt's climate change impact and government adaptation strategies, emphasizing the urgent need for action to avoid catastrophic consequences like rising sea levels, droughts, famine, and biodiversity loss. The definition of "climate change" lacks international consensus, leading to varying interpretations. In Egypt, situated in Northeast Africa with coasts on the Mediterranean and Red Seas, the climate is predominantly dry and hot, with a warming trend observed over recent decades. The country's geography and economy are deeply intertwined with the Nile River, agriculture being a significant contributor to GDP despite challenges like inefficient irrigation and land degradation. Industrial growth, particularly in energy-intensive sectors like cement and fertilizers, adds to greenhouse gas emissions. Social challenges persist, with poverty and urbanization exacerbating issues such as informal settlements encroaching on agricultural land. Egypt's vulnerability to climate change is recognized globally,

Fig 2

with projections indicating potential threats to its development and poverty reduction efforts.

Impacts of climate change:

1- **Agriculture:** USAID assists smallholder farmers in Upper Egypt and the Delta, addressing climate change challenges like extreme temperatures and water scarcity. They provide technical support for crop production and post-harvest processes, promoting climate-smart practices such as composting farm waste instead of burning it. Additionally, USAID offers low-cost solutions to boost crop resilience and water efficiency, while also focusing on gender-specific impacts and improving nutrition knowledge among female farmers.

2- **Water:** Since the 1970s, USAID has invested $3.5 billion to improve access to clean water and sanitation for over 25 million Egyptians, reducing child mortality by 80%. Their efforts enhance water utilities, increasing operational efficiency and financial sustainability. With USAID support, water losses have decreased by 8%, and energy usage has been optimized. Additionally, USAID aids in shaping water policies, updating water sector education, and funding research to address water-related challenges.

3- **Renewable Energy:** USAID is assisting Egypt in achieving its target of 42% renewable energy capacity by 2030 by fostering a supportive regulatory and financial landscape for clean energy investments. Collaborating with the government and private sector, USAID introduced the New and Renewable Energy Diploma Program, equipping technical school students with skills necessary for the renewable energy sector. Additionally, USAID influences energy policy, enhances energy education, and funds innovative research to tackle climate change. To broaden access to clean energy, USAID facilitates loans for small-scale farmers to adopt solar panels for irrigation instead of diesel fuel. In FY 2022, USAID allocated $10 million to Egypt's Nexus for Water, Food, and Energy Initiative (NWFE), aiding in the deployment of 10 gigawatts of renewable energy capacity by providing technical support and enhancing the enabling environment for large-scale renewable energy projects.

And all these very dangerous problems can have unbelievable effects on the Egyptian community. Egypt seeks to solve all the problems.

Increase the industrial and agricultural bases of Egypt.

Fig 3

Industry and agriculture are the most important Economic activities in Egypt, and they also have a great role in the national income. As a result of the presence of a variety of natural resources, industry, and agriculture are the most common economic activities in Egypt, so many people depend on it as a main source of income. according to this, many of the resources have been consumed and this affects the economy of Egypt. Consequently, the income of many people gets affected by the decrease in the number of resources and they have to work on another activity, as a result, the problem has become worse.

So, the country started to develop new methods and new projects that depend on new technologies to develop these activities and avoid the negative consequences of the deterioration of the economy. All of these methods aim for sustainability. To develop agriculture, they use strategies that aim to the following to solve the problem of agriculture.

Fig 4

1. Improving irrigation systems, reaching the maximum economic effectiveness of allocating and using land and water resources, and ensuring a growth rate of agricultural output by about 3.6% per year.

2. Achieving high levels of food security of strategic commodities, strengthening the competitiveness of agricultural products in the world's markets, and going towards environmentally friendly organic farming and safe use of pesticides to improve the quality of agricultural products.

3. Allocating direct investments to implement supportive programs for the agriculture sector,

including the expansion of the area of agricultural land, and mechanization of the system of agricultural tenure, with around EGP 87.5 million allocated to complete the project of the mechanization system,

4. Building a database of holders and providing audited data to help follow the agricultural development programs.

Also, in the industrial field, the country started to build projects that depended on new scientific methods and technologies to develop the industry. Also, they started to use sustainable energy sources as a solution to the problem of pollution and the high cost of unsustainable energy sources.

Finally, the country is on the way to a modern scientific sustainable economy.

Problem to be solved.

Global warming and climate change are important problems. Global warming and climate change are a result of the increase in the concentration of greenhouse gases in the atmosphere. Greenhouse gases are gases in Earth's atmosphere that trap heat, leading to the greenhouse effect. They allow sunlight to enter the atmosphere freely but prevent some of the outgoing infrared radiation from escaping back into space, effectively trapping heat within the Earth's atmosphere. This process warms the planet and is essential for maintaining temperatures conducive to life, but human activities have significantly increased the concentrations of these gases, leading to global warming and climate change. Industrial building materials have a big role in increasing the concentration of greenhouse gases because manufacturing of these materials leads to the emission of greenhouse gases (CO2, CH4, and N2o). The mission or the challenge is to solve this problem. The whole world is trying to solve this problem. One of the solutions is to use natural building materials instead of using industrial materials. Using natural building materials can have several positive impacts on climate change:

1. Carbon Sequestration: Many natural building materials, such as wood, straw, and clay, can sequester carbon dioxide from the atmosphere during their growth or production. For example, trees absorb CO2 as they grow, and when harvested sustainably, wood can store carbon for the life of the building. This helps to reduce the concentration of greenhouse gases in the atmosphere, mitigating climate change.

2. Low Embodied Energy: Natural building materials often have lower embodied energy compared to conventional materials like concrete and steel. Embodied energy refers to the total energy required to extract, process, manufacture, transport, and install a material. Since natural materials are often locally sourced and minimally processed, they have a lower carbon footprint associated with their production, leading to reduced greenhouse gas emissions.

3. Renewable and Sustainable: Many natural building materials are renewable and can be sourced from sustainably managed forests, agricultural lands, or natural deposits. This reduces dependence on finite resources and minimizes environmental degradation associated with extraction and manufacturing

processes. Additionally, using renewable materials supports local economies and promotes sustainable land management practices.

4. Energy Efficiency: Natural materials such as straw, clay, and rammed earth offer excellent thermal insulation properties, helping to regulate indoor temperatures and reduce the need for artificial heating and cooling. This can lead to lower energy consumption over the lifetime of the building, resulting in fewer greenhouse gas emissions associated with energy production.

5. Improved Indoor Air Quality: Natural materials are often non-toxic and emit fewer volatile organic compounds (VOCs) compared to synthetic materials like plastics and foams. This can contribute to better indoor air quality and occupant health, reducing the need for ventilation systems that consume energy and emit greenhouse gases.

The failure to address climate change could have severe and far-reaching negative impacts on both human societies and the natural world. Some of the consequences of not solving the problem of climate change include:

1. Extreme Weather Events: Climate change is expected to increase the frequency and intensity of extreme weather events such as hurricanes, droughts, heat waves, and heavy rainfall. These events can lead to loss of life, destruction of infrastructure, displacement of populations, and economic disruption.

2. Rising Sea Levels: As global temperatures rise, glaciers and polar ice caps melt, contributing to a rise in sea levels. This poses a significant threat to coastal communities, resulting in increased flooding, erosion, saltwater intrusion into freshwater sources, and loss of valuable coastal habitats.

3. Impacts on Agriculture: Changes in temperature and precipitation patterns can disrupt agricultural systems, leading to reduced crop yields, decreased food security, and increased vulnerability to pests, diseases, and invasive species. This can exacerbate hunger, malnutrition, and poverty, particularly in developing countries.

4. Loss of Biodiversity: Climate change threatens the survival of many species and ecosystems, leading to loss of biodiversity and ecosystem services. This can have cascading effects on ecosystems, including disruptions to pollination, nutrient cycling, and water purification processes, as well as increased risk of species extinction.

5. Health Risks: Climate change can have direct and indirect impacts on human health, including heat-related illnesses, respiratory problems from air pollution, waterborne diseases, and the spread of vector-borne diseases such as malaria and dengue fever. Vulnerable populations, including the elderly, children, and those with pre-existing health conditions, are particularly at risk.

6. Social and Economic Disparities: Climate change can exacerbate existing social and economic inequalities, disproportionately affecting marginalized communities and exacerbating poverty, inequality, and social unrest. This can lead to increased conflict over scarce resources, migration pressures, and challenges to governance and social cohesion.

7. Loss of Livelihoods: Many livelihoods and industries, particularly those dependent on natural resources such as agriculture, fisheries, forestry, and tourism, are vulnerable to the impacts of climate change. Disruptions to these sectors can lead to loss of income, unemployment, and economic instability, particularly in vulnerable regions.

8. Infrastructure Damage: Climate change can damage critical infrastructure such as roads, bridges, buildings, and utilities, leading to increased maintenance costs, disruptions to services, and decreased resilience to future climate impacts.

Design and implement a sustainable, secure, and technologically advanced home that addresses key challenges such as smoke leakage detection, high-security measures, and efficient temperature control. The solution should integrate eco-friendly materials, smart home automation, and innovative sensor technologies to ensure optimal comfort, safety, and energy efficiency for inhabitants while minimizing environmental impact.

1-Smoke Sensors: These sensors can detect the presence of smoke in the air and trigger an alarm or notification then send it to the Microcontroller (e.g., Arduino or Raspberry Pi): The microcontroller processes data from the smoke sensors and triggers actions such as activating ventilation systems or sending alerts then take the action by the Relays or Motor Drivers: These components can be used to control ventilation systems or activate buzzers and hoods when smoke is detected, and Wi-Fi or Bluetooth Module: Enables communication with smartphones to send notifications to homeowners in real-time, and make a thing, and reliable and timely

detection and response to smoke leakage events, ensuring the safety and well-being of occupants and minimizing property damage.

2-High Security with IR Sensors: Infrared (IR) Motion Sensors: These sensors detect motion within their detection range and can be placed on doors and specific rooms to monitor movement. Microcontroller: Similar to the smoke detection system, a microcontroller processes data from the IR sensors and triggers actions such as sending notifications or activating lights, LED Lights, or Smart Bulbs: Lights can be turned on automatically upon detecting motion, enhancing security and providing visibility, Smartphone Integration: Allows homeowners to receive notifications when motion is detected and remotely monitor their home's security status, importance: IR sensors enhance home security by detecting unauthorized entry or movement within the house. Real-time notifications and automated lighting deter potential intruders and provide homeowners with immediate awareness of security breaches, ensuring the safety of their property and loved ones.

3-Temperature Sensors: These sensors measure indoor and outdoor temperatures, providing data for temperature control systems.

RESEARCH

- What is the definition of a smart home?
- How to solve the problem of climate change?
- Climate change, and people its causes, and its impacts.
- How to reduce the effect of climate change?
- what are the best building materials in thermal insulation?
- What are the most effective sensors?
- How to decrease the temperature?
- Prior solution of smart buildings.

PRIOR SOLUTIONS

BENGALURU smart city

Abstract

The concept, origin, development, and history of Smart Cities were studied in conjunction with the backdrop, town, and economic characteristics of Bengaluru. The idea of Smart Cities is a "booming" global phenomenon, with Smart City initiatives being implemented worldwide. After the 2000s, the term "smart" gained popularity in urban policymaking to describe the skillful use of information technologies (IT) to improve a city's infrastructure and service functionality and efficacy. With significant contributions from various technologies, including computer science, information technology, remote sensing, and enhanced multimedia, the idea of Smart Cities is a recent one in our country. This project explores the foundation of a Smart City by examining existing relevant parameters, such as electricity, solid waste management systems, water supply, and transportation, in Kondashettihalli and Chikkaballapur areas only due to the vastness of the city and the deciding parameters. Bengaluru's spatial expansion is examined, the shortcomings of the current situation are contrasted with those of a Smart City, and lastly, planning plans and recommendations are provided.

Key Words: Information technology; Smart city; Spatial Growth; Transportation; Water supply.

Introduction

Although the idea of a "smart city" is not new, the term came from the "smart growth" movement at the close of the 1990s, which promoted new guidelines for urban planning (Harrison).

Fig 5

see Donnelly (2011). With more than 2000 Smart City projects launched in Europe, Asia, Africa, and America, the concept of smart cities is a rapidly expanding global phenomenon. More than 1500 projects were in progress by 2013, growing at a 20% compound annual growth rate. Following the turn of the millennium, the term "smart" found widespread usage in urban policymaking to refer to the astute use of information technology (IT) to boost the efficiency and productivity of a city's infrastructure and services (Robin Hodkinson 2011).

The components of the "smart city" phenomenon are explained in detail below. With significant assistance from several technologies, such as information technology, computer science, remote sensing, and improved multimedia, the concept of a "smart city" is new in our nation. The effectiveness, agility, adaptability, and sustainability of Smart City operations are all enhanced by these new solutions. GIFT (Gujarat International Finance Tec-City), India's first Smart City, was founded in Gujarat in 2011 with help from CISCO and ABB specialists. Haldia (West Bengal), Amanora (Maharashtra), Pune, Lavasa (Maharashtra), and Electronic City (Bangalore) are some more Indian cities that have Smart City initiatives.

Definition of Smart City

A smart city is a term used to describe a city that utilizes information and communication technologies (ICT) to enhance the efficiency, performance, and overall well-being of its residents. The goal of a smart city is to integrate technology into various aspects of urban life, such as infrastructure, transportation, healthcare, education, and governance, to improve the quality of services, reduce resource consumption, and enhance the overall sustainability of the city.

Physical Characteristics of The City

Bangalore is situated in the Indian state of Karnataka in the southeast. It is located at an average elevation of 900 meters in the Mysore Plateau, which is a portion of the broader Precambrian Deccan Plateau.

2,953 feet. Situated at 12.97°N 77.56°E, it spans a 741 km2 area. While the surrounding rural areas are included in the Bangalore Rural district, the majority of Bangalore is located inside the Bangalore Urban district of Karnataka. The Karnataka government created the new district of Ramanagara from the old Bangalore Rural district. Bangalore's topography is mostly flat, with a few hilly patches in the western regions. The tallest peak is Vidyaranyapura Doddabettahalli, which is situated northwest of the city at a height of 962 meters (3,156 feet). The Arkavathi and South Pennar rivers meet 60 kilometers north of the city in the Nandi Hills, although no major rivers flow through the city itself. The river Vishabhavathi, a small tributary of the Arkavathi that flows through the city, originates near Basavanagudi. A large portion of Bangalore's sewage is shared by the Arkavathi and Vrishabhavathi rivers.

Five sewage Bangalore's 1922-constructed, 215 km2 sewage infrastructure is connected to treatment facilities.

of the urban area.

The Madivala Tank, Hebbal Lake, Ulsoor Lake, Yediyur Lake, and Sankey Tank are Bangalore's largest freshwater lakes and tanks. Layers of alluvial sediments, varying from silty to sandy, contain groundwater. The Peninsular Gneissic Complex (PGC), which is composed of granites, gneiss, and migmatites, is the most common type of rock in the area.

Red, fine loamy to clayey soils and red laterite make up Bangalore's soil composition. A few coconut trees and sizable deciduous canopies make up the majority of the city's vegetation. Bangalore is located in seismic zone II, which is a stable zone, although it has nevertheless seen earthquakes as strong as 4.5.

Steps involved in the Research work (i) A survey of the key research articles that have influenced the area of sustainable smart cities. (ii) Review of the Most Important Policy Literature, Including Initiatives, Experiences, and Recommendations from the World's Major Organizations and Experts in the Subject. (iii) Building a repository of such efforts through the process of gathering documentation of Smart City ideas as well as the execution performance. (iv) To assist in planning, implementation, and assessment of smart city ideas, create a framework based on contributions derived out of case study development, research, and policy literature review; (v) Due to time restrictions, four parameters - transportation, water supply, solid waste management, and electricity, are taken into consideration. (vi) Data collection and analysis: (a) Revised Master Plan 2015 (b) BBMP Solid Waste management system data of the required areas (c) Land use in 2004 (d) Existing land use 2015 (e) Final proposals: After analyzing all the data shortfall is calculated, issues and strategies are found, and the final proposal is prepared.

Guidelines and Features for Smart City the Smart Communities Mission seeks to assist communities that provide a solid foundation for their citizens, a high standard of life, a healthy, sustainable environment, and the use of "Smart" Solutions. The goal is to analyze compact geographic areas and create a replicable model that would serve as a beacon for other aspirational communities. Sustainable and equitable development

are the main priorities. A report, according to the Ministry of Urban Development, Government of India, 2015, some common characteristics of the extensive growth of smart cities are listed below: (i) Encouraging mixed-use development in localized areas - planning for "unplanned regions" that have a variety of appropriate applications and activities close to one another to maximize the use of available land. The States will allow some wiggle room in their land use and building bylaws to accommodate change. (ii) Inclusivity and housing- increase access to homes for everybody. (iii) Creating walkable localities - increases the local economy, encourages interactions, and provides security while reducing traffic, air pollution, and resource depletion. All crucial administrative tasks are located nearby, and roads are constructed or refurbished not only for automobiles and public transportation but also for pedestrians and cyclists. (iv) Preserving and developing open spaces - playgrounds, parks, and recreational areas to improve the

standard of living for residents, lessen the effects of urban heat in some areas, and overall promote eco-balance. (v) Encouraging a range of transportation choices - Public transportation, last-mile connectivity, and transit-oriented development (TOD) for paratransit. (vi) Making government more cost and citizen-friendly - There is a growing reliance on online services to promote accountability and transparency, particularly when using mobile devices to lower service costs and deliver services without requiring residents to visit municipal offices; setting up online focus groups to hear from the public and gather feedback; and using online monitoring of programs and activities with the help of virtual site visits. (vii) Giving the city an identity- based on its primary economic activity, such as its regional cuisine, education, sporting goods, arts and crafts, culture, health, furniture, hosiery, textiles, dairy products, etc. (viii) Improving area-based development's infrastructure and services through the application of smart solutions. Using less energy, fewer resources, and offering less expensive services are a few examples.

Analysis of Existing Land Use Data, 2015

Land use type	Area (Hectare)
Industrial	1518940.631
Commercial	1098771.663
Residential	2193468.931
Park and Open space	208997.6717
Civic Amenities	346366.8219
Transport	2487435.916
Total	**7853981.634**

**Land Use 2015 data is obtained from Revised Master Plan (RMP) 2015

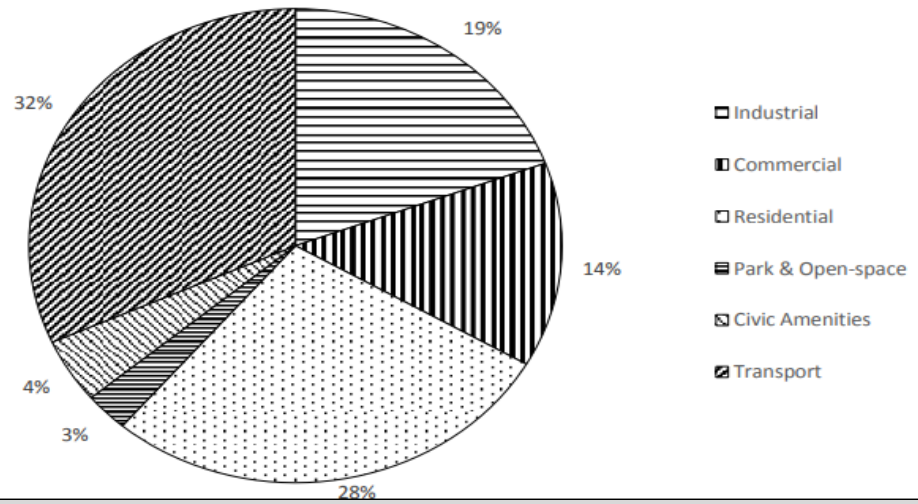


Table 1

Conclusion

Transportation, social infrastructure, stormwater drainage, sewage, solid waste management, electricity, telecommunication, and water supply are the cornerstones of a smart city. Only four pillars have been examined and planning solutions have been suggested for them because of the size of the study region. To create a more succinct plan for the creation of a smart city, more research can be done on the remaining pillars. It is projected that land usage for parks, open spaces, commercial and civic facilities, and residential will decrease while land use for transportation and residential and parkland will increase based on the spatial growth analyses conducted in 2004 and 2015. Reducing commercial land and increasing industrial and open spaces are necessary for both environmental sustainability and economic prosperity.

The proposed infrastructure should be built with transportation facilities in mind. The suggested plans and prompt monitoring should be used to check for

deficiencies in the solid waste management system. Water usage and losses in the water supply segment

ought to be governed in line with the suggested ideas. One important aspect of urban activities is an uninterrupted power supply, which can be accomplished using the suggested harvesting technique. The planning ideas and strategies are meant to fill in the gaps left by the chosen smart city pillars. As a result, a basic plan for all smart city initiatives is created.

Nile City Towers

we can discuss the types of smart technologies commonly integrated into modern mixed-use developments like Nile City Towers:

Fig 6

1. **Building Management System (BMS):** A centralized Building Management System could be implemented to monitor and control various building systems and subsystems within Nile City Towers. This includes HVAC (heating, ventilation, and air conditioning), lighting, security, fire safety, and energy management systems. The BMS provides real-time data analytics, remote monitoring, and automated control to optimize building operations, enhance energy efficiency, and ensure occupant comfort.
2. **Smart Lighting Controls:** Nile City Towers may feature smart lighting systems equipped with sensors and controls that adjust lighting levels based on occupancy, natural light availability, and time of day. Adaptive lighting controls can help conserve energy, reduce electricity costs, and create a comfortable and productive environment for occupants.
3. **Energy Management Systems:** Advanced energy management systems could be deployed to monitor, analyze, and optimize energy usage within Nile City Towers. This includes tracking energy consumption patterns, identifying energy-saving opportunities, and implementing strategies to reduce energy waste and operating costs. Real-time energy monitoring dashboards and automated energy-saving measures help achieve sustainability goals and enhance the building's environmental performance.

4. **Occupancy Sensors and Smart HVAC Controls**: Occupancy sensors installed throughout Nile City Towers detect human presence in different areas and adjust HVAC settings accordingly. Smart HVAC controls regulate temperature, airflow, and ventilation based on occupancy patterns and usage requirements, ensuring optimal indoor air quality and thermal comfort while minimizing energy consumption.
5. **Integrated Security and Access Control: Advanced** security systems, including surveillance cameras, access control systems, and intrusion detection sensors, are essential for ensuring the safety and security of Nile City Towers. Integrated security management platforms provide centralized monitoring, real-time alerts, and automated responses to security incidents, enhancing the overall security posture of the development.
6. **Smart Parking Management:** Nile City Towers may incorporate smart parking management systems to optimize parking availability, enhance traffic flow, and improve the overall parking experience for tenants, visitors, and guests. This includes real-time parking occupancy monitoring, digital signage for guidance, and mobile apps for reservations and payments, reducing congestion and enhancing convenience.
7. **Digital Signage and Communication Systems:** Digital signage displays, and communication systems installed throughout Nile City Towers provide information, announcements, and emergency alerts to occupants and visitors. Interactive touchscreens, wayfinding kiosks, and mobile apps enhance communication, navigation, and engagement within the development.

The Edge, Amsterdam Showcasing an exemplary IoT building.

The challenge: Modern office buildings require considerable flexibility to accommodate new uses and working patterns, and also need to meet high standards of sustainability. When Deloitte Netherlands began envisaging their new Amsterdam office, they not only considered the working environment but also the natural environment. That is to say, the building should have a double emphasis on productivity and sustainability. Moreover, the office would be extremely energy efficient with a strong emphasis on workplace efficiency. The new building is planned to house the company's 1,700 local employees (previously spread over several buildings) in one modern space, creating a visible symbol of the company's digital approach and contemporary outlook. Working in the digital age differs from traditional office life – it is much more flexible, decentralized, and collaborative. In the digital age, employees are no longer desk-bound, or even office-bound; they might choose to work from home (at least a couple of days a week) and will go to the office mainly for meetings and social interaction with their colleagues. As such, it is not uncommon that 50% of desks in a conventional office are unoccupied at any one-time during office hours – a waste of urban space, and a rallying call for a radically different building design and configuration of rooms. This trend towards greater flexibility affects other types of construction too, of course. Transport infrastructure, for example, has to adjust to the growing numbers and changing tastes of travelers. Heathrow Terminal 5 in London was specifically designed as an

Fig 7

adaptable building that can be easily transformed internally to cater to different demands. As for the other dimension, sustainability, it is no longer a mere optional add-on offered by property developers, but a key feature of new buildings, actively promoted by developers and demanded by owners, tenants, and end-users. The sustainable design is not just to comply with regulatory requirements, such as energy efficiency specifications or renewable-energy policies, but also out of genuine environmental concern.

The idea: Combine smart building design and innovative technologies to improve sustainability, workforce interaction, and end-user experience. Developed by OVG Real Estate and designed by PLP Architecture, The Edge was completed in November 2014. In its 40,000 square meters of floor space, it sets new standards in sustainability, technology, flexibility, and user experience, and shows how smart building design can reduce energy demands. The most prominent architectural feature is the multi-story, north-facing glass atrium, which admits abundant daylight while on the south façade, concrete walls absorb heat (and shield the interior from sunlight), and solar panels on that façade and the roof convert that sunlight into energy. Perhaps the most innovative element of The Edge is the way that the many different elements of the building are interconnected, pushing the boundary of the Internet of Things. Every technical system in the building is controlled within a single network, enabling a live view of the building and modulated for maximum efficiency: the lift, the lighting and cooling systems, the robot that cruises around the building at night as a security guard, even the coffee machines and towel dispensers. By working closely with suppliers, the developer introduced 21 innovations that had never been applied before, such as Philips' Ethernet-connected lighting and a safe plastic for cable insulations that is non-toxic in

Fig 8

the event of fire. These and other solutions, according to Erik Ubels, Chief Technology Officer at OVG, are “integrated on the silicon level, instead of just gluing technologies together”. Consider the 6,000 low-energy LED luminaires, on which the lighting system is based: they contain multiple sensors for measuring temperature and light, and even for detecting movement. These sensors – and a further 28,000 positioned throughout the building – are linked to the linked to the building network and integrated into a data analytics platform enabling smart facility management. This helps, for example, guide cleaning staff to heavily used areas, or refill the towel dispenser before it runs out, and allows predictive maintenance of the LED lights by monitoring their usage.

John Deere Operations Center

1-Telematics and GPS Technology:

John Deere equipment is equipped with telematics devices that gather data on various parameters, including location, fuel usage, and engine performance.

2-Field and Crop Monitoring:

Utilizing sensors and GPS technology, farmers can monitor field conditions, track planting patterns, and assess crop health in real time.

3-Data Analytics:

The Operations Center employs advanced analytics to process data collected from equipment and sensors. It provides insights into field performance, crop variability, and yield potential.

4-Weather Integration:

The platform integrates weather data to help farmers make informed decisions based on current and forecasted weather conditions. This includes planning for irrigation, harvesting, and addressing potential weather-related challenges.

5-Automated Documentation:

The system automates the documentation process, recording details such as planting dates, fertilizer applications, and crop yields. This data helps farmers analyze trends and optimize future practices.

6-Remote Equipment Management:

Farmers can remotely monitor and manage their equipment through the Operations Center. This includes tracking the location of machinery, monitoring fuel levels, and receiving alerts for maintenance needs.

7-Rate Technology (VRT):

VRT allows for the precise application of inputs such as seeds, fertilizers, and pesticides based on the specific needs of different areas within a field. This optimization leads to resource efficiency and improved yields.

8-Mobile App Access:

Farmers can access the Operations Center through a mobile app, providing flexibility and real-time insights while they are in the field.

Benefits:

Precision Farming: The system enables precision farming practices, optimizing resource usage and improving overall efficiency.

Data-driven decision-making: Farmers can make informed decisions based on real-time data, leading to improved crop management and increased productivity.

Remote Monitoring: The ability to remotely monitor equipment and field conditions allows for timely interventions and reduces the need for physical presence on the farm.

Sustainability: By optimizing inputs and reducing waste, the system contributes to more sustainable agricultural practices.

The John Deere Operations Center is an excellent example of how a Smart Agriculture System can be integrated into the existing workflows of farmers, providing them with valuable insights and tools to enhance productivity and sustainability.

Living sustainably

is becoming increasingly crucial as we work to lessen our environmental impact and protect natural resources for coming generations.

Numerous easy and efficient methods exist to make your home more sustainable, and this is a fantastic place to start.

There are numerous methods to live a more sustainable lifestyle at home, such as growing your food, using sustainable products, and consuming less energy.

and this is an example of a sustainable house.

Fig 9

Text description provided by the architects. Occupying one of four plots forming a gap in a typical Brixton terrace, Slip House constitutes a new prototype for adaptable terraced housing. Three simple ‘slipped’ orthogonal box forms break up the bulk of the building and give it its striking sculptural quality.

The top floor is clad in milky, translucent glass planks, which continue past the roof deck to create a high-level ‘sky garden’. Designed to Code for Sustainable Homes Level 5, it features ‘energy piles’ utilizing a solar-assisted ground source heat pump creating a thermal store beneath the building. PV’s, a wildflower roof, rainwater harvesting, reduced water consumption, mechanical ventilation with heat recovery within an airtight envelope with massive levels of insulation make this one of the most energy-efficient houses built in the UK. A prototype brownfield development offering dense, flexible, urban living – the house is a vehicle for in-house research into sustainable design, seamlessly integrating the often-conflicting aesthetic requirements of architecture and alternative low-energy systems. We are working to develop this model for multiple developments and affordable housing.

Living and working (‘Living over the shop’) is something that interests us. We see a prototype new ‘terraced’ house, squeezed into underutilized city (Brownfield) sites. This flexible type of home can allow for the artisan or homeworker to sub-let or downsize. This can enliven local communities and produce ‘homes’ that create opportunities rather than be dormitories or financial assets. Slip House is flexible and can be used as a single home, studio workspace, and apartment, or two apartments.

The perimeter walls are loading-bearing, freeing up the internal areas of supporting columns or additional load-bearing walls. The house's open-plan layout ensures that walls/dividers are simple to erect and require minimal construction effort. This aspect of Slip House is not only financially sustainable but also environmentally so, as it helps to ensure the permanence of the overall structure, as minimal modifications can allow the house to adapt to changing lives and living situations indefinitely.

Our approach was to model the building as a series of simple orthogonal box forms that use the full width of the site. This allows future buildings to simply adjoin the flank walls.

The house takes the idea of three slipped boxes. The boxes are carefully placed to maximize light and outlook from the inside while not intruding on the neighbor's outlook. The shifting planes also break up the bulk of the building and give it its sculptural quality.

Design requirements.

1. Deal with the existence of any smoke leakage BY:
    - Opening all windows to help get out the smoke.
    - Turning on buzzers and hoods.
    - Sensing a message to the chosen cell phone.
2. High security by Installing IR sensor on the door of the home and on specific rooms inside it to make some actions:
    - Send a message to a chosen cell phone.
    - On entering a specific room in the home turn on the light and send a message to the owner.
3. Decide the temperature from the outside at least 3 degrees from the inside.

selection of solution

After determining design requirements, we started to think about an idea and choose suitable materials. Our solution divides into two parts. The first one is to choose building materials for the smart home, and the second part is to determine the smart system of this home and its components. First, we will build this house mainly with clay. We will use it in a durable form, and then it can be left to completely get dry. The advantage of clay is that it's energy efficient, which is the main purpose of the project, as it can store heat and make it cooler indoors. This is one of our design requirements. And it's natural material. We will reeds as reinforcement for the structure of the house. Reeds are also natural and cannot be broken easily so we can guarantee a strong house. The second part of the solution is the smart system of the house and its components. First, we will use ESP32. This is a micro controller that act as the precursor to the whole system. We will use two sensors: LM sensor which is a temperature sensor. IR sensor which will act as motion sensor. It's responsible for recording any object that passes along the sensor, and as feedback there will be a lead that will illuminate when any object passes, and we will use a buzzer. We will use a server motor to control motion of an object such as a door. And we used DC geared motor. These motors act as actuators. This is the description of the solution that will meet design requirements.

Clay (Building material)

ESB 32 (Micro controller)

LM sensor (temperature sensor)

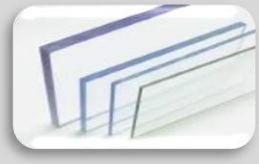
SELECTION OF PROTOTYPE

The design for our prototype is pretty much like a cylinder placed vertically in running water (Ex: river), there will be a removed portion of the cylinder vertically along the cylinder to let the passage of the water, this opening will not be located in the center of the cylinder, it will be located on a side to give the turbine the ability to rotate, then there will be a larger hole opposite to the first hole to let the water flow out Our prototype has the same design that European homes have. If you want to imagine it, its cross section is like a triangle that is above a square because this inclined surface of the sides of the roof is important to make the required angle for the solar panels to generate clean energy. The building materials of the prototype are (clay and palm branches) These materials are natural and have the capability to make good thermal insulation, because we aim to decrease the temperature inside the building 3C using building materials only.

Fig 10

Items	Quantity	Description	Usage	Cost	source	Image
Palm branches	100	Palm branches	The main material of the home.	100 L. E	Palm farms	
Pottery clay	1 kg	a natural material created by weathered rock. It is soft, malleable, and will permanently harden if baked at high temperatures, making it a practical material for making tableware	The main material of the home.	70 L. E	the countryside	
Arduino Uno	1	Arduino Uno is a microcontroller board based on the ATmega328P (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs)	To control the sensors	513 L. E	electronics shop	
MQ-2	1	This smoke sensor module with an MQ2 Smoke sensor has both digital and analog outputs, you also can adjust the sensitivity with the potentiometer	Smoke detector	80 L. E	electronics shop	

12V pc fan	1	DC 12V 4 Pin Black Plastic PC Cooling Fan 90mm x 90mm	For ventilation	130 L. E	electronics shop	
HC05	1	H C-05 is a Bluetooth module that is designed for wireless communication. This module can be used in a master or slave configuration.	To send data	250 L. E	electronics shop	
Servo	2	Micro Servo Motor (1.3 kg. cm)	To open and close windows	95 L. E	electronics shop	

IR sensor	1	An IR sensor typically consists of an IR emitter (usually an IR LED) and an IR receiver (usually a photodiode or phototransistor).	To discover movement	35 L. E	electronics shop	
LM35	2	The LM35 series are precision IC temperature sensors, with an output voltage linearly proportional to the Centigrade temperature	To measure temperature	75 L. E	electronics shop	
polycarbonate	1	is a transparent thermoplastic with carbonate functional groups.	To make windows	50 L. E	Glass shop	

keypad	1	Self-adhesive membrane keypad 4×4 – 16 keys. Diaphragm keypad equipped with 16 keys in a 4 x 4 arrangement. It has numbers: from 0 to 9, letters: A, B	To enter numbers	50 L.E	electronics shop	
buzzer	1	This is a round buzzer that operates around the audible 2 kHz range. You can drive it directly from a 5V PIC or Arduino to generate the tones	Discovering data from the sensor	10 L.E	electronics shop	

Table (2)

METHODS

Building and constructing the home itself:

We make some holes in the branch to collect and fix the rest.

We make each side separately then collect the ropes.

We used isolated polycarbonic sheets to construct the windows and doors.

We fixed the hinges on the windows and door.

We cover our home with clay pottery to decrease the temperature inside the home.

Connecting components and coding:

We searched for the connections for the components. And started to make these connections in our project.

We started to test every sensor separately to be sure of its functionality.

We started to write the code for each system using IDE, and then we developed them.

We finally made the test plan for the whole system and building.

TEST PLAN

First, we charge our batteries with solar cells via a charging module. when the batteries become fully charged. We used it to run the Arduino. We uploaded our code to Arduino and tested it with some steps.

1. The user entered a certain code (111) on the keypad, which is located outside the door, to activate security and entered another code to deactivate it (333).

• The security system worked as follows: by putting something near the IR sensor The IR gets a signal, and depending on it, a buzzer will trigger, and a warning notification will be sent to the user.

2. We check the MQ2 sensor. and the sensor gets a fast response and runs the fan to get rid of the smoke or gas and the servo motor to open the window. In addition, there is a warning message to the user, the LED will turn on, and the buzzer will trigger with another tone.

3. We get a read of the internal temperature sensor and the external temperature sensor and represent them in the application.

Data Collection

Cuboid		Volume
length	60 ± 1 cm	72000 ± 1 cm^3
width	30 ± 1 cm	
height	40 ± 1 cm	

Table. 1 (cuboid dimensions)

Triangle prism	Area of base	Height	Volume
	1800 ± 1 cm^2	18.5 ± 1 cm	33300 ± 1 cm^3

Table. 2 (prism dimensions)

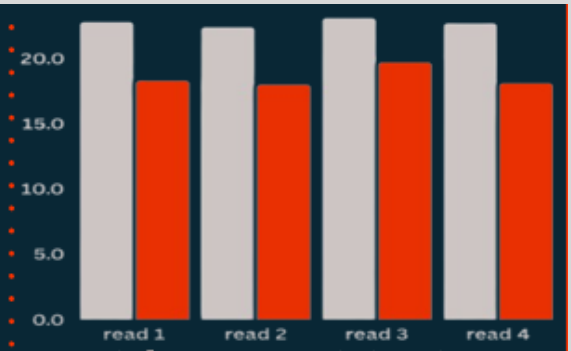


fig. 2 (temperature reading in Celsius)

	Read 1	Read 2	Read 3	Read 4
Inside	18.3	18.0	19.7	18.1
outside	22.8	22.4	23.1	22.7

fig. 3 (temperature reading in Celsius)

ANALYSIS

Our project is divided into some partitions: Building the home: We chose some materials that are efficient in isolating the temperature; these materials are palm branches, which will be covered with pottery clay. Natural shade from palm branches can drastically cut down on the amount of direct sunlight that reaches the ground or neighboring structures. Strong winds can be repelled by using palm branches as a natural barrier. They can lower the ambient temperature by obstructing or decelerating air circulation, which can produce pockets of colder air behind them. Because pottery clay has a high thermal mass, heat energy may be efficiently absorbed and stored in it. The clay takes in heat from the surroundings during the day while it's hot outside. This heat is released by the clay at night when

the temperature drops, assisting in the maintenance of a more constant interior temperature. Because of its capacity to both absorb and release moisture from the air, clay can help control the humidity levels within buildings. By using pottery clay, you can use fewer synthetic materials, which have a greater impact on the environment. Choosing the suitable components for the circuit: Our home consists of three main systems: the first is a smoke leakage system based on using an MQ-2 smoke sensor to detect any leakage, and accordingly, some instructions are going to be executed (opening the upper window using a servo, turning on a fan to extract air, turning on a buzzer as an alert, blinking a LED, and sending a message to a selected cellphone). The second is a security system using an IR sensor. This idea is based on arming the system by entering a specific code on the keypad, and while the system is activating, the IR sensor detects any motion inside the home, triggers an alert, and sends a message to a cellphone saying that there is a stranger in the home. To disarm the system, another code must be printed on the keypad. The third is using two temperature sensors to measure the temperature outside and inside the home and send the readings to a cellphone. The power supply for the system is: We used solar cells as a source of electricity to charge the battery and use it to power the Arduino. A 30S-20A lithium-ion battery charger is used.

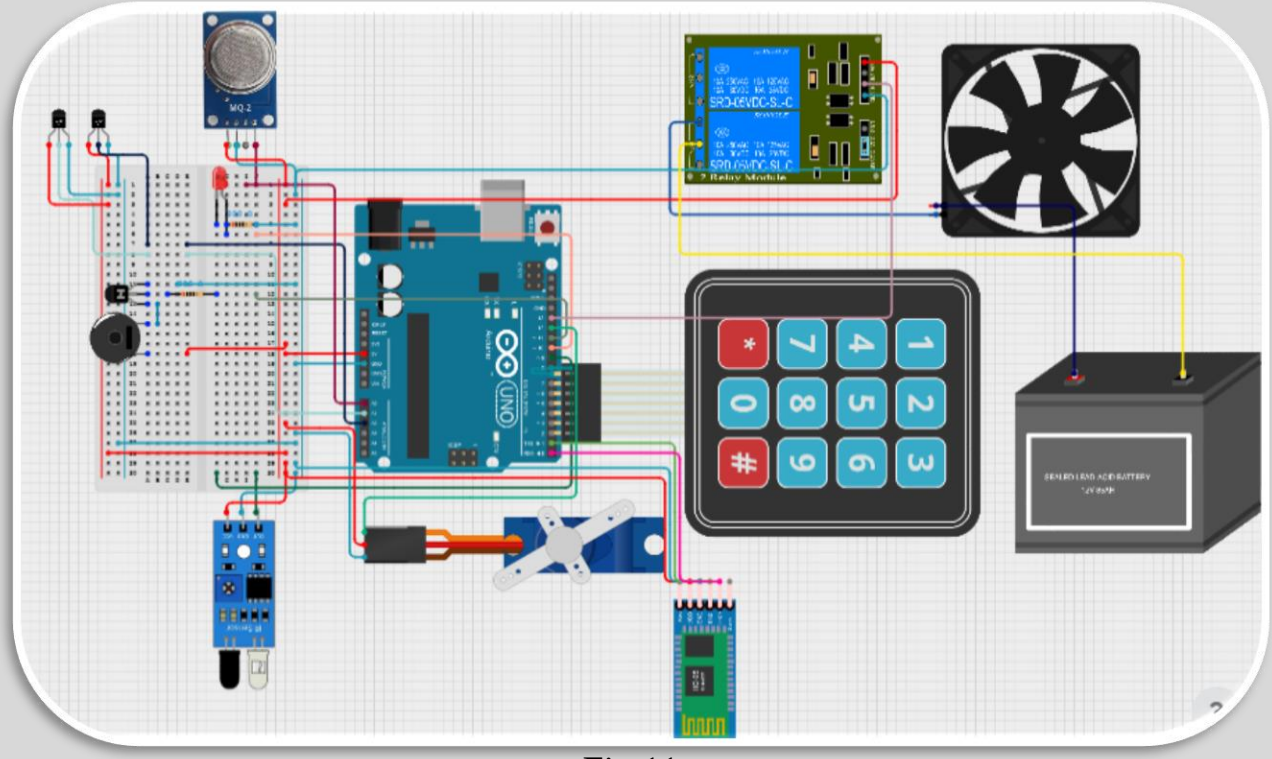


Fig 11

```
// Read the analog values from the sensors
int sensorValue1 = analogRead(tempinside);
int sensorValue2 = analogRead(tempoutside);

// Convert analog values to temperature in Celsius
float mv1 = (sensorValue1 /1024.0) *500; // LM35 output is 10 mV per degree Celsius
float mv2 = (sensorValue2 /1024.0) *500;
float temperature1 = mv1/10;
  float temperature2 = mv2/10;

// Print the temperatures on the serial monitor
Serial.print("Temperature Sensor outside: ");
Serial.print(temperature1);
Serial.println(" °C");

Serial.print("Temperature Sensor inside: ");
Serial.print(temperature2);
Serial.println(" °C");
// Send temperatures over Bluetooth
bluetooth.print("Temperature outside: ");
bluetooth.print(temperature1);
bluetooth.println(" °C");

bluetooth.print("Temperature inside: ");
bluetooth.print(temperature2);
bluetooth.println(" °C");
// Wait for a short delay before reading the sensors again
delay(1000);
```

Fig (12): temperature code

This is the code for the temperature sensor to determine the inside and outside temperatures. The variables are defined for the temperature outside and inside the home. sensor Value/1024.0: If the sensor has a 10-bit ADC (Analog-to-Digital Converter) resolution, 1024 is the maximum value, and this normalizes the sensor data to a number between 0 and 1. The normalized value is scaled by multiplying it by 500 to match the output of the LM35 sensor, which produces 10 mV per degree Celsius. Therefore, the normalized value is effectively converted to millivolts by multiplying by 500. float temperature1 = mv1/10: Since the LM35 sensor outputs 10 mV per degree Celsius, dividing mv1 by 10 gives the temperature in degrees Celsius. Bluetooth. Println (temperature): print the values of the temperature in the cell phone using the HC05 Bluetooth module.

```
int sensorValue = analogRead(mq2Pin); // Read analog value from MQ-2 sensor
if (sensorValue > 350) { // If sensor reading exceeds the threshold
  bluetooth.println("smoke detected"); // Send "Gas detected" over Bluetooth
  Serial.println("smoke detected"); // Print "Gas detected" to serial monitor for debugging
  digitalWrite(fan,HIGH);
  myservo.Write(180);
  digitalWrite(led, HIGH);
  digitalWrite(BUZZER_PIN, HIGH); // Turn on the LED indicator
  delay(5000); // Wait for 10 seconds
  digitalWrite(led, LOW);
  digitalWrite(BUZZER_PIN, LOW);// Turn off the LED indicator
   digitalWrite(fan,LOW);
  myservo.Write(0);
}
else {

  digitalWrite(led, LOW);
  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(fan,LOW);
  myservo.Write(0);
}
```

Fig (13): smoke leakage system

The analog value from the MQ-2 sensor attached to pin mq2Pin is read using the formula **int sensor Value = analog Read(mq2Pin)**. The voltage measured by the sensor is returned by the analog Read() function as a value between 0 and 1023. **if (sensor Value > 350)** { Verifies whether the sensor reading is higher than the 350-threshold value. **Bluetooth. println("smoke detected");** Bluetooth transmits the

alert "smoke detected." This is probably used to tell another device like a smartphone about the detection.

Activates an LED and buzzer with the code **digital Write(led, HIGH), digital Write(buzzer, HIGH).** And then turning them off if there is no smoke. Also opens the window using the servo by **my servo. Write(180)** and turn on a fan by **digital Write(fan, HIGH),** and turn them off when there is no smoke.

```
void armSystem() {
  if (authenticateUser("111")) {
    systemArmed = true;
    digitalWrite(BUZZER_PIN, HIGH);  // Turn on the buzzer to indicate armed state
    delay(1000); // Delay for one second
    digitalWrite(BUZZER_PIN, LOW); // Turn off the buzzer after one second
    bluetooth.println("System armed");
  } else {
    bluetooth.println("Incorrect arm code");
  }
}

void disarmSystem() {
  if (authenticateUser("333")) {
    systemArmed = false;
    bluetooth.println("System disarmed");
  } else {
    bluetooth.println("Incorrect disarm code");
  }
}
```

Fig (14): security system

Arming the System: The user enters a specific code using the keypad to arm the system. Once the correct code is entered, the system activates the IR sensor to detect any motion.

Disarm the System: The user enters another specific code to disarm the system. Upon successful entry of the disarm code, the system deactivates the IR sensor.

Intrusion Detection: While the system is armed, if the IR sensor detects motion, it triggers the buzzer to sound an alarm, indicating a potential intrusion.

Functions:

- arm System(): It checks if the user enters the correct code (in this case, "111"). If authenticated, it sets the system Armed to true, activates the

buzzer briefly, and sends a message via Bluetooth indicating that the system is armed.

- disarm System(): It checks if the user enters the correct disarm code (in this case, "333"). If authenticated, it sets the system Armed to false and sends a message via Bluetooth indicating that the system is disarmed.
- authenticate User(String code): This function prompts the user to enter a code via the keypad until the # key is pressed. It then compares the entered code with the provided one and returns true if they match.
- send Alert(): This function sends a message via Bluetooth and activates the buzzer briefly to alert about a detected stranger (motion detected by the IR sensor).

RECOMMENDATIONS

While building the prototype, we faced problems with each smart system and the house's structure. I think avoiding these problems can save time and money. Here are some recommendations that can help in making a better project:

1. The first source of information is workshops or experts not only websites or web sources.

2. It’s better to buy new electronic materials, use them for the first time, and work efficiently.

3. Building the house with more adhesive material such as concrete that can carry heavy weights and help in inserting the system inside the house

LEARNING OUTCOMES

BI.2.09	Understanding the concept of feedback control system through more and more learning about the hormonal control and feedback control system in the human reproductive system
MA.07	we have studied the concept of derivatives and how to use it to measure the suitable slope of the solar panels above the roof to generate electricity
Cs.2.05	we have studied the concept of loops (feedback) in programming languages, where a certain action is repeated if a certain condition is true.
EN. W	We studied the different types of writing for example scientific writing and academic writing and how to differentiate between these types.
BI.2.10	We have studied different theories like evolution theory and how Darwin followed the scientific method to apply his theory
ph.2.08	Study Faraday's law and know more about induction and its types of self or mutual
Geo.2.09	Study the evolution of Earth and its layers that enable us to know materials that are stable in high-temperature
Ph.2.10	Study AC generators, DC generators, and DC motors. To know how can we control the velocity and the power of the actuators we are going to use
Ch.2.09	Study different types of batteries and how they work in charging and discharging and knowing their components.
ENG.2.01	We learned in English about mind mapping

Table (3): Learning Outcomes

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